
Position: Model Collapse Does Not Mean What You Think

Rylan Schaeffer¹ Joshua Kazdan² Alvan Caleb Arulandu³ Sanmi Koyejo¹

Abstract

The proliferation of AI-generated content online has fueled concerns over *model collapse*, a degradation in future generative models’ performance when trained on synthetic data generated by earlier models. Industry leaders, premier research journals and popular science publications alike have prophesied catastrophic societal consequences stemming from model collapse. In this position piece, we contend this widespread narrative fundamentally misunderstands the scientific evidence. We highlight that research on model collapse actually encompasses eight distinct and at times conflicting definitions of model collapse, and argue that inconsistent terminology within and between papers has hindered building a comprehensive understanding of model collapse. To assess how significantly different interpretations of model collapse threaten future generative models, we posit what we believe are realistic conditions for studying model collapse and then conduct a rigorous assessment of the literature’s methodologies through this lens. While we leave room for reasonable disagreement, our analysis of research studies, weighted by how faithfully each study matches real-world conditions, leads us to conclude that certain predicted claims of model collapse rely on assumptions and conditions that poorly match real-world conditions, and in fact several prominent collapse scenarios are readily avoidable. Altogether, this position paper argues that model collapse has been warped from a nuanced multifaceted consideration into an oversimplified threat, and that the evidence suggests specific harms more likely under society’s current trajectory have received disproportionately less attention.

¹Stanford Computer Science ²Stanford Statistics
³Harvard University. Correspondence to: Rylan Schaeffer
 <rschaeff@cs.stanford.edu>, Sanmi Koyejo
 <sanmi@cs.stanford.edu>.

1. Introduction

The rapid surge of AI-generated content has sparked intense debate about potential ramifications of training future generative AI models on datasets containing synthetic data generated by previous models. One especially concerning prediction is *model collapse*: a phenomenon whereby future generative models fail due to being trained on synthetic data. Model collapse has captured attention at the highest levels of academia and industry: *Nature* prominently featured model collapse in 2024 (Gibney, 2024) alongside accompanying research suggesting that AI models trained on synthetic data would suffer catastrophic degradation in performance (Shumailov et al., 2024), while prominent science and news outlets like *Scientific American* and the *Wall Street Journal* amplified these concerns, writing “a training diet of AI-generated text, even in small quantities, eventually becomes poisonous to the model being trained.” (Rao, 2023) and that “feeding a model text that is itself generated by AI is considered the computer-science version of inbreeding” (Seetharaman, 2024). Meanwhile, some industry leaders have highlighted model collapse as a critical challenge for the future of AI development and deployment (Wang, 2024).

In this position piece, we argue that this widespread narrative of model collapse, which describes a bleak future filled with polluted pretraining data and useless generative models, oversimplifies or misinterprets both the precise scientific claims and their underlying assumptions and mechanisms. Through careful analysis, we identify three critical gaps between the prevailing discourse and research reality:

First, we reveal that the term “model collapse” encompasses eight definitions of performance degradation in model-data feedback loops. This multiplicity of definitions, used inconsistently between papers and at times inconsistently within papers, has resulted in papers talking past one another, thereby hindering development of a comprehensive understanding of likely futures for frontier deep generative models. We argue specific failure modes should be explicitly identified and discussed alongside comparable results.

Second, we posit trends that we believe faithfully describe common practices of leading AI labs pretraining frontier AI systems on web-scale data: increasing compute, improving data quality, and expanding datasets of real and synthetic

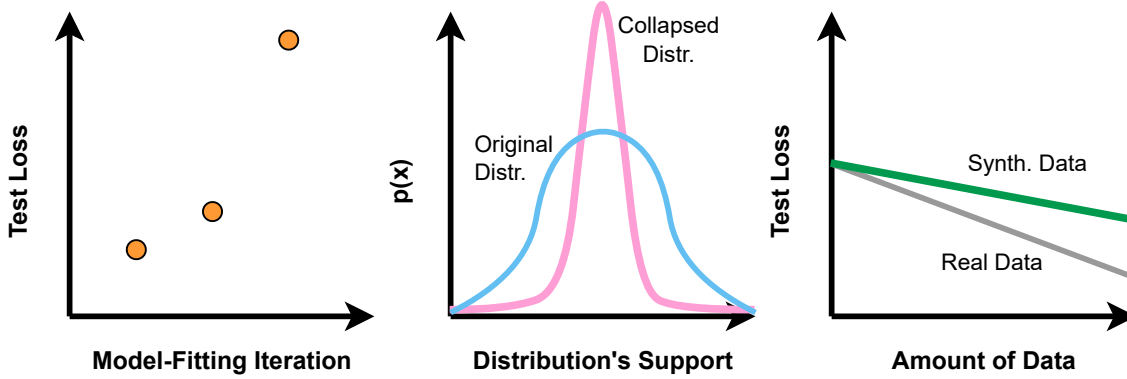


Figure 1. Model Collapse Has Been Defined in Multiple and Sometimes Conflicting Ways. By hand-annotating 28 prior research publications, we identify 8 definitions of model collapse (Sec. 2). The 8 definitions can be loosely grouped into three families: (1) the behavior of the test loss on real data over model-fitting iterations (left), (2) the deformation of the real data distribution over model-fitting iterations (center) and (3) the scaling behavior of the test loss with respect to typical scaling quantities such as the amount of data (right).

data. One key point that we emphasize here is that many prominent model collapse papers assume data are entirely deleted after each model-fitting iteration and that subsequent models are trained entirely on synthetic data generated by their predecessors, which we argue is not realistic.

Third, we weigh different results in the model collapse literature based on the plausibility of their assumptions along multiple axes of consideration to assess which notions of model collapse pose significant and likely threats to future frontier AI models. We argue that some definitions of model collapse do not correspond to catastrophic outcomes under our realistic assumptions, and most concerning collapse predictions emerge from implausible experimental setups. However, there are very real threats to tails of the data distribution that should be taken seriously.

Altogether we argue that model collapse has been inflated from a precise and important technical consideration into a mischaracterized and overstated threat. While synthetic data poses genuine challenges that warrant careful study, our analysis reveals that the most widely stated collapse scenarios can be avoided through standard ongoing practices in model development and dataset curation. Instead of worrying about unrealistic catastrophic notions of model collapse, by adopting a more realistic perspective, we can re-orient to focus on the real issues of diversity collapse happening now (Zhang et al., 2024; Padmakumar & He, 2024; Murthy et al., 2024; Wu et al., 2024). This position paper aims to clarify the scientific discourse around model collapse, propose best practices for future work on the subject, and redirect research attention towards understanding how to generate and curate synthetic data that improves future frontier AI systems while mitigating failure modes.

2. Definitions of Model Collapse

We begin with a non-obvious but critical point: the model collapse literature has at least eight different definitions based on different notions of model performance degradation. As evidence, we hand-annotated twenty-eight prominent prior research publications on model collapse to determine which papers offer explicit definition(s) of model collapse (Fig. 2), where we define explicit as either (1) any mathematical definition, or (2) any precise verbal description of the failure behavior *independent* from results. We additionally categorize which definition(s) of model collapse each paper uses, perhaps implicitly, across any and all mathematical and empirical results (Fig. 2 bottom). We find that many papers do not offer explicit definitions of model collapse and also sometimes use multiple definitions, leading to a lack of specificity and apparent contradictions both within single works and across multiple works.

Before delving into the eight definitions, we first define some shared terminology. In general, we consider model-data feedback loops whereby f_t is the t -th generative model, and we study the behavior of a sequence of generative models $(f_t)_t$ that are iteratively fit to data and then sampled from. We call data sampled from a generative model *synthetic data*. We can evaluate the quality of a generative model in multiple ways. One prominent way is via the *population risk*, defined as the expected loss over the entire real data distribution $\mathbb{E}_{x \sim P}[\ell(f_t(x))]$, where x is some real datum, P is the real data distribution, and ℓ is the loss function (Vapnik, 1991). Another way to evaluate the quality of a generative model is by its *tail risk*, defined as the expected loss conditioned on tail events $\mathbb{E}_{x \in \text{Tail}(P)}[\ell(f_t(x))]$, where $\text{Tail}(P)$ informally represents real data with low probability.

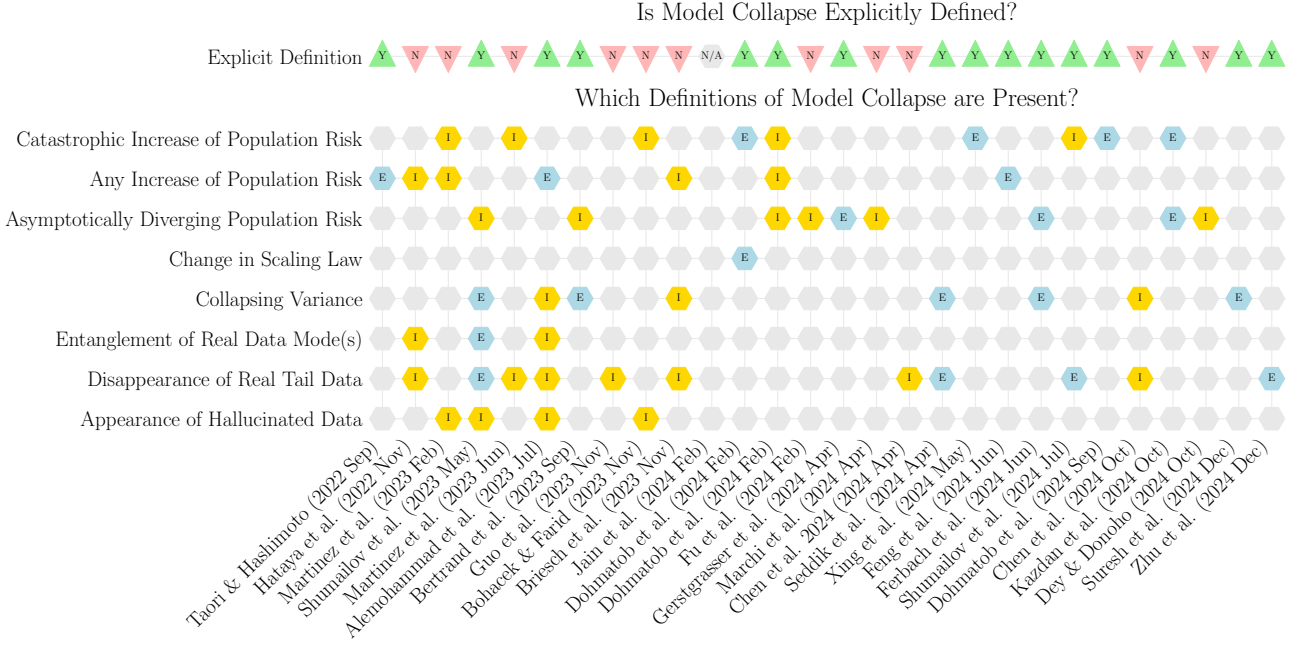


Figure 2. **Model Collapse has been defined in multiple and sometimes conflicting ways.** We conduct a meta-analysis of research papers on model collapse. Top: We identify which papers offer *any* explicit definition of model collapse (**Yes** (Y) or **No** (N)), broadly construed. Bottom: We identify which definition(s) of model collapse each paper uses for its experimental and/or mathematical results, either **explicitly** (E) or **implicitly** (I). Our annotations reveal that research on model collapse is based on multiple definitions that we will show sometimes conflict between papers and even within individual papers.

Population risk provides a holistic view of performance, but may mask specific failure modes (Xu, 2024; Kozerawski et al., 2022), whereas tail risk can reveal degradations in edge cases even when population risk remains stable (Hoffmann & Börner, 2020; Xu, 2024). There are many other salient properties of generative models one can use to assess whether the models collapse.

1. **Catastrophic Increase of Population Risk** (Dohmatob et al., 2024b; Bertrand et al., 2023; Kazdan et al., 2024b): Perhaps the most colloquial definition, model collapse is a critical and rapid degradation in model performance due to the presence of synthetic data, as measured by population risk. We note that what constitutes catastrophic is often undefined.
2. **Any Increase of Population Risk** (Alemohammad et al., 2023; Dohmatob et al., 2024b): Under this strict definition, model collapse occurs if there is *any* increase in population risk when training with synthetic data compared to training with real data alone.
3. **Asymptotically Diverging Population Risk** (Gerstgrasser et al., 2024; Kazdan et al., 2024b; Dey & Donoho, 2024): This definition considers model collapse to occur when the population risk grows without

bound over successive model-fitting iterations. This represents a fundamentally unstable learning dynamic where each iteration of synthetic data generation and training leads to progressively worse performance.

4. **Collapsing Variance** (Alemohammad et al., 2023; Shumailov et al., 2023; Bertrand et al., 2023) Model collapse here is when variance (or diversity) trends towards 0 and the learned distributions tend towards delta-like functions over successive model-fitting iterations.
5. **Change in Scaling Law** (Dohmatob et al., 2024c): In this view, model collapse occurs if the governing scaling behavior changes due to the presence of synthetic data. Specifically, model collapse occurs if the relationship between model performance and training data size deviates from the expected scaling behavior observed with real data.
6. **Disappearance of or Entanglement of Real Data Mode(s)** (Alemohammad et al., 2023): Sometimes called “Mode Collapse” (Goodfellow et al., 2014; Lucic et al., 2018; Brock et al., 2019), model collapse here is defined by the presence of synthetic data preventing the model from learning particular modes of